National University of Computer and Emerging Sciences
Karachi Campus

Artificial Intelligence (AI2002)

Final Exam

Date: May 27th 2024

Course Instructor(s)

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Total Time (Hrs): 3
Total Marks: 50
Total Questions: 7

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Roll No Section Student Signature

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Attempt all the questions.

CLO#1: To grasp the concepts of rational behavior and intelligent agents

Q1: Answer the following brief questions

1. What are the primary characteristics that define a rational agent in Artificial Intelligence?
2. Why CSPs use factored state representation? Give justification.
3. An autonomous drone delivery system operates in a complex environment where it must deliver packages to various locations within a city. Explain whether the environment for the agent is episodic or sequential. Justify your answer.
4. How does local beam search differ from standard hill-climbing algorithms in terms of handling multiple states simultaneously during search?
5. What are the key differences between depth-first search (DFS) and breadth-first search (BFS) in terms of space and time complexity?
6. Assume that a rook can move on a chessboard any number of squares in a straight line, vertically or horizontally, but cannot jump over other pieces. Manhattan distance is an admissible heuristic for the problem of moving the rook from square A to square B in the smallest number of moves. (T/F). Justify.
7. Minimax algorithm leads to outcomes at least as good as any other strategy only when the opponent is also optimal. Give Justification.
8. Write any two limitations of hill climbing algorithm in optimization problems
9. Given a CSP with n variables of domain size d, what would be the size of the search tree: (i) with commutativity (ii) Without commutativity.
10. Iterative deepening search can also be applied to adversarial search. Give Justification. **[10 Marks]**

CLO#3: To showcase comprehension and capability in implementing key concepts and methodologies in probabilistic reasoning, machine learning algorithms, and neural networks

Q2: (A). Consider a population in which 30% of individuals have a certain genetic trait, denoted as Trait A. Among those with Trait A, 70% also have Trait B, while among those without Trait A, 20% have Trait B.

1. What is the conditional probability that an individual has Trait B given that they have Trait A?
2. What is the unconditional probability that a randomly selected individual from the population has Trait B?

(B). Calculate the Joint Probability Distribution of the following figure 1.

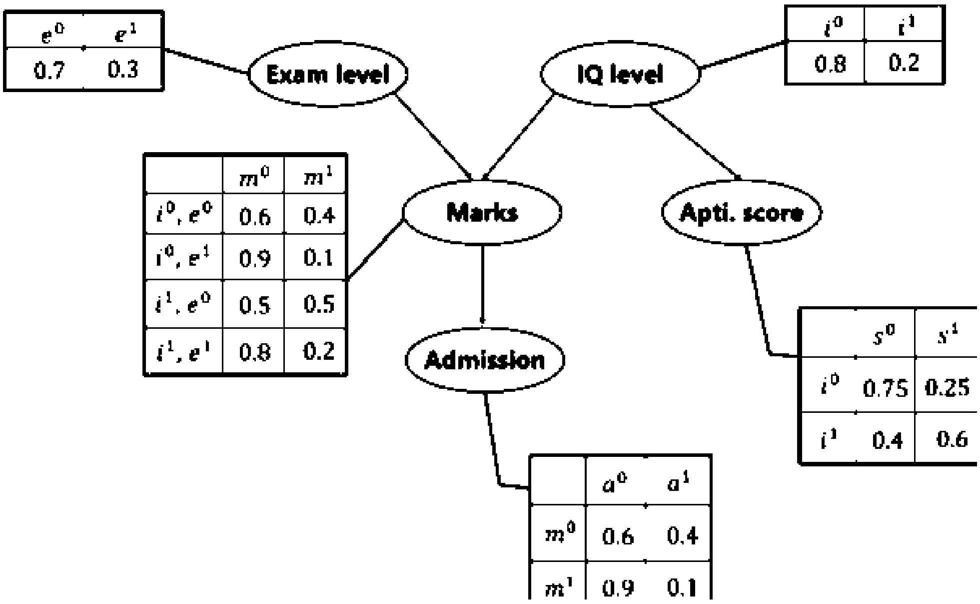

Figure 1: Bayesian Network

1. Calculate the probability that in spite of the exam level being difficult, the student having a low IQ level and a low Aptitude Score, manages to pass the exam and secure admission to the university.
2. In another case, calculate the probability that the student has a High IQ level and Aptitude Score, the exam being easy yet fails to pass and does not secure admission to the university.

[2+2+2 = 6 Marks]

Q3: (A). How does Naïve Bayes handle feature independence and missing data?

(B). Build a Naïve Bayes Classifier for the given table 1 and use your model to classify the following table 2 as new instances.

Instance	Education Level	Career	Years of Experience	Salary
1	High School	Management	Less than 3	Low
2	High School	Management	3 to 10	Low
3	College	Management	Less than 3	High
4	College	Service	More than 10	Low
5	High School	Service	3 to 10	Low
6	College	Service	3 to 10	High
7	College	Management	More than 10	High
8	College	Service	Less than 3	Low
9	High School	Management	More than 10	High
10	High School	Service	More than 10	Low

Table 1: Training Data

Instance	Education Level	Career	Years of Experience
1	High School	Service	Less than 3
2	College	Retail	Less than 3
3	Graduate	Service	3 to 10

Table 2: Test Data

[2 + 4 = 6 Marks]

Q4: (A). List down two applications of regression and correlation.

(B). The yield of a chemical process is a random variable whose value is considered to be a linear function of the temperature x. The following data in table 3 of corresponding values of x and y.

Temperature in C° (x)	0	25	50	75	100	125
Yield in Grams (y)	10	25	40	55	70	?

Table 3: Training Data

Find the estimated yield in grams using regression, if the temperature is raised to 125 C°

[1 + 3 = 4 Marks]

Q5: (A). You are a data analyst at a healthcare organization. Your task is to predict whether a patient is likely to have diabetes based on historical patient data. The dataset includes the following features for each patient:
- Age - BMI (Body Mass Index) - Blood Pressure - Insulin Level - Glucose Level
- Diabetes status (1 for diabetes, 0 for no diabetes)

Given the following data in table 4, use the K-Nearest Neighbors (KNN) algorithm to classify whether a new patient with the following characteristics will have diabetes, use k=3:

Age:	BMI:	Blood Pressure	Insulin Level	Glucose Level
45	28	80	100	140

Table 4: Test Data

The historical data is in table 5 as follows:

Age	BMI	Blood Pressure	Insulin Level	Glucose Level	Diabetes
25	22	70	85	130	0
30	30	85	90	150	1
35	28	80	105	160	1
40	25	75	100	140	0
50	27	78	95	145	0
55	31	90	110	155	1
60	26	85	108	148	0
65	29	88	115	165	1
70	32	92	120	170	1

Table 5: Training Data

Tasks:
1. Calculate the Euclidean distance between the new patient and all patients in the dataset.
2. Identify the K-nearest neighbors based on the distances calculated.
3. Predict whether the new patient will have diabetes using majority voting.
4. Discuss how the choice of k affects the classification result and how you might select an optimal k.

[1.5 + 1 + 1 + 0.5 = 4 Marks]

Q5: (B). Construct a complete decision tree using the ID3 algorithm on the provided data. Display the entire tree from the root node to leaf nodes as indicated in the set. Dataset of animal classification have the following attributes:

- Habitat (Forest, Desert, Grassland)
- Diet (Herbivore, Carnivore, Omnivore)
- Activity (Diurnal, Nocturnal)

Animal Type (Mammal, Bird, Reptile) - this is the target variable. **[8 Marks]**

Habitat	Diet	Activity	Animal Type
Forest	Herbivore	Diurnal	Mammal
Forest	Carnivore	Nocturnal	Mammal
Desert	Carnivore	Nocturnal	Reptile
Grassland	Herbivore	Diurnal	Bird
Forest	Omnivore	Nocturnal	Mammal
Grassland	Omnivore	Diurnal	Bird
Desert	Omnivore	Nocturnal	Reptile
Grassland	Herbivore	Diurnal	Bird
Forest	Herbivore	Nocturnal	Mammal
Desert	Herbivore	Diurnal	Reptile

Table 6: Training Data

Q6: Use the k-means algorithm and Euclidean distance to cluster the following 8 examples into 3 clusters:

Data Points	A1	A2	A3	A4	A5	A6	A7	A8
x	2	2	8	5	7	6	1	4

Suppose that the initial seeds (centers of each cluster) are A1, A4 and A7. Initially, run the k-means algorithm for 2 epochs. After the 2^{nd} epoch, please give the following:

a) New Centroid Values
b) Newly formed cluster sets.
c) How many more iterations are needed to converge? Please Justify. **[2+2+2= 6 Marks]**

Q7: Consider a feedforward neural network with one input layer consisting of two input neurons, two hidden layers with three neurons each, and one output layer with one neuron. The activation function used in all layers is the sigmoid function. Draw the neural network and compute the output.

Input Layer to First Hidden Layer (Layer 1):	First Hidden Layer to Second Hidden Layer (Layer 2):	Second Hidden Layer to Output Layer (Layer 3):
Neuron 1: Weight 1: 0.1, Bias: -0.2 Weight 2: -0.3, Bias: 0.4 Neuron 2: Weight 1: 0.5, Bias: -0.1 Weight 2: -0.2, Bias: 0.3 Neuron 3: Weight 1: 0.4, Bias: 0.2 Weight 2: -0.1, Bias: -0.5	Neuron 1: Weight 1: 0.2, Bias: 0.1 Weight 2: -0.4, Bias: 0.3 Weight 3: 0.5, Bias: -0.2 Neuron 2: Weight 1: 0.3, Bias: -0.4 Weight 2: -0.2, Bias: 0.2 Weight 3: 0.1, Bias: 0.3 Neuron 3: Weight 1: -0.1, Bias: 0.2 Weight 2: 0.4, Bias: -0.3 Weight 3: -0.2, Bias: 0.1	Neuron 1: Weight 1: -0.3, Bias: 0.5 Weight 2: 0.2, Bias: -0.1 Weight 3: 0.4, Bias: 0.3

Given an input vector [0.6, -0.1], calculate the output value of the neural network. Assume the sigmoid activation function for all neurons. **[6 Marks]**